Lam-Nguyen Nguyen

Nguyễn Lâm Nguyên

Undergraduate Researcher · Vinh Long University of Technology Education (VLUTE), Vietnam

Email: lamnguyen.it04@gmail.com · GitHub: github.com/lawnguyenit · Location: Vinh Long, Vietnam

RESEARCH INTERESTS

Trustworthy Machine Learning for IoT/Edge Systems, Weak Supervision, Temporal ML Evaluation, AI Security.

EDUCATION

Vinh Long University of Technology Education **2022 – Present**

B.S. Candidate in Information Technology, Cybersecurity Track

Cumulative GPA: **3.19 / 4.00**

PUBLICATIONS & MANUSCRIPTS

Benchmarking Foundation Models and Gradient Boosting for Agricultural IoT Context Analytics: A Weakly Supervised Temporal Evaluation **2026**

Hoai-Hanh Tran, Thanh-Nghi Doan, ***Lam-Nguyen Nguyen***

Manuscript submitted to EAI AICON 2026.

Observation Contracts and Effective Evidence in Temporal Rule-Derived Agricultural IoT Targets **2026**

Lam-Nguyen Nguyen*, Hoai-Hanh Tran, Thanh-Nghi Doan*

First-author manuscript in preparation.

RESEARCH EXPERIENCE

First-Author Research — Trustworthy ML for Agricultural IoT **2026 – Present**

- Designed controlled experiments to evaluate rule-derived weak targets under different temporal information regimes and deployment conditions.
- Investigated target leakage, proxy features, temporal history, distribution shift, and the distinction between label recoverability and independent physical validation.

Undergraduate Thesis Researcher — AgriFusion-IoT **2025 – 2026**

Advisor: Tran Hoai Hanh · Project: github.com/lawnguyenit/AgriFusion-IoT

- Designed and deployed an ESP32-S3/4G agricultural IoT system and built a traceable pipeline from raw Firebase telemetry to weakly labeled ML benchmarks using temporal splits and leakage controls.

Undergraduate Researcher — Agricultural IoT Analytics **2026**

- Contributed to data preparation, weak labeling, temporal evaluation protocols, and benchmarking tabular models against gradient boosting for conference manuscript submitted to EAI AICON 2026.

HONORS & AWARDS

Honorable Mention, Algebra Division B — National Student Mathematics Olympiad **2025**

University Representative, National Student Mathematics Olympiad **4 Editions (2023–2026)**

TECHNICAL SKILLS

Programming: Python (data processing, ML experimentation), C/C++ (ESP32/embedded development).

ML & Data: XGBoost, scikit-learn, tabular classification, rule-derived weak labeling, time-based train/test evaluation, imbalanced classification metrics.

IoT & Embedded: ESP32-S3, RS485/Modbus, SHT30, SIM A7682S (4G LTE), Firebase Realtime Database.

Development: Git/GitHub, Linux.

LANGUAGES

English: CEFR B1 (institutional certificate, VLUTE) · **Vietnamese:** Native.